



Paper ID : 260917

Printed Page: 1 of 1

Subject Code : BMC054

Roll No:

MCA

(SEM. IV) THEORY EXAMINATION 2025-26

MACHINE LEARNING

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

(a)	Define machine learning. Mention any two real-life applications of machine learning.	CO1	K2
(b)	What is a well-defined learning problem? State its three components.	CO1	K2
(c)	Define overfitting in machine learning.	CO2	K1
(d)	State Bayes theorem and mention its use in classification.	CO2	K1
(e)	What is information gain in decision tree learning?	CO2	K1
(f)	Define deep learning and mention one of the latest trends in machine learning.	CO3	K2
(g)	What is meant by model optimization? Mention one measure used to report expected accuracy.	CO5	K2

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	Explain the types of learning in machine learning. Differentiate supervised, unsupervised and reinforcement learning with suitable examples.	CO1	K2
(b)	Apply linear regression for prediction and explain the role of the cost function and gradient descent in model learning.	CO2	K3
(c)	Discuss the latest trends in machine learning, such as deep learning, reinforcement learning and genetic algorithms, with suitable applications.	CO3	K3
(d)	Design a k-nearest neighbour-based classifier for a real-world classification problem. Discuss the role of k and distance measures in prediction.	CO4	K4
(e)	Explain model evaluation using accuracy, precision, recall and F1-score. Why is accuracy alone not sufficient for imbalanced data?	CO5	K5

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain the need for machine learning for problem-solving. Also, describe the general steps involved in designing a learning system.	CO1	K2
(b)	Discuss the history and major approaches of machine learning, including artificial neural networks, clustering, decision tree learning, Bayesian networks, SVM and genetic algorithm.	CO1	K2

4. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain logistic regression. How is it different from linear regression? Describe its use in binary classification.	CO2	K3
(b)	Explain Support Vector Machine. Discuss hyperplane, decision surface, support vectors and different kernel functions used in SVM.	CO2	K3

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Discuss recent trends in machine learning. Explain how deep learning and convolutional neural networks are used in real-world applications.	CO3	K3
(b)	Explain reinforcement learning and deep Q-learning as the latest trends in machine learning. Discuss their practical applications.	CO3	K3

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Design an artificial neural network model for a classification problem. Discuss the role of perceptron, multilayer perceptron, activation function, weights and bias.	CO4	K4
(b)	Design a convolutional neural network for an image classification problem. Explain the convolutional layer, activation function, pooling layer and fully connected layer with a suitable case study.	CO4	K4

7. Attempt any one part of the following:

7 x 1 = 7

(a)	Explain model optimization in machine learning. Discuss how accuracy, precision, recall and F1-score can be used to report the expected performance of a learned model.	CO5	K5
(b)	Explain the Markov Decision Process and the Q-learning algorithm. How can Q-values be optimized to improve the learned policy and expected accuracy?	CO5	K5